



Department of Information Technology

Academic Year 2024 - 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / Equivalence of pushdown automata and CFG-CFG to PDA
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO9
- **Activity Chosen:** Learning by Teaching
- **Justification:**

In unit III, the students need to understand the Equivalence of pushdown automata and CFG-CFG to PDA. This activity supports the students to understand the voluntarism and leadership.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - A topic Equivalence of pushdown automata and CFG-CFG to PDA was given to Ms. Divya Sri J S two days before the presentation.
 - She prepared and taken the class on 19.04.2025.
 - The topic was delivered around 10 minutes first she discussed the equivalence of CFG and then she discussed the equivalence of CFG and PDA with example.
 - The implementation of Learning by teaching activity as shown in figure1 and figure2.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO3	3	3	3	1	2	1	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO3	3	3	3	1	2	1	2	1	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of Theory of Computation for Equivalence of pushdown automata and CFG-CFG to PDA	The students will be able to identify the need for analysis of complex engineering problems	Able to design the process flow to prove Equivalence of pushdown automata and CFG-CFG to PDA	Able to analyze the data to prove Equivalence of pushdown automata and CFG-CFG to PDA	The students will be able to follow ethics during the proof	The students able to function as a team for converting Equivalence of pushdown automata and CFG-CFG to PDA	The students will be able to communicate one another and can able to present	The students able to learn and implement the concept in Natural Language Processing	The students able to apply mathematical and computing skills.	The students can Exhibit knowledge of languages in the field of Cloud Computing to develop IT solutions	The students able to exhibit knowledge in Languages to develop a quality product.

- **Images / Screenshot of the practice:**



Figure 1: Learning by Teaching Handled by Divya sri JS

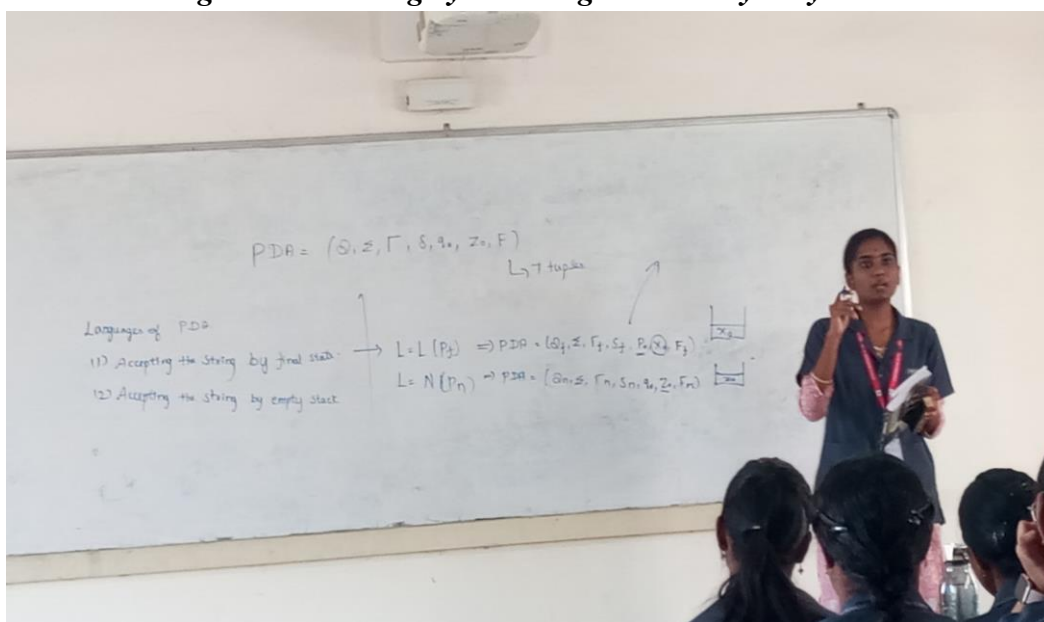


Figure 2: Learning by Teaching

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Each student was interested in listening to the class and understood the importance of the equivalence between CFG and PDA.
- The students grasped the concept of the equivalence theorem.
- The session empowered students to gain knowledge about the equivalence of Context-Free Grammars (CFG) and Pushdown Automata (PDA).

- ❖ **Benefit of the practice:**

- Students possess the ability to analyze Context-Free Grammars (CFGs) and Pushdown Automata (PDAs).
- Students understand the significance of CFGs and PDAs.
- This activity helps students recognize the importance of CFGs and PDAs.

❖ *Challenges faced in implementation:*

- Some students feel difficult to understand the topic. After presentation summary of the topic given by the faculty.

References:

- ❖ <https://www.teaching-learning.utas.edu.au/learning-activities-and-delivery-modes/planning-learning-activities/examples-of-learning-activities>.
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/>

Signature of Faculty Member

HOD